

Problem

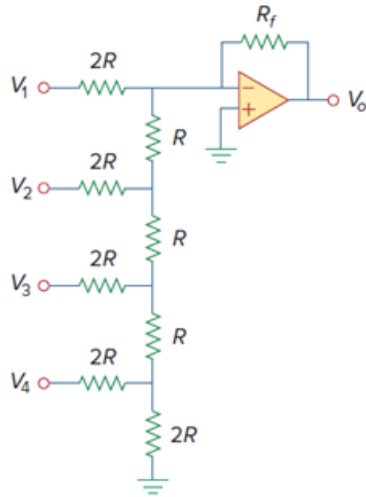
A four-bit R - $2R$ ladder DAC is presented in Fig. 5.103.

(a) Show that the output voltage is given by

$$-V_o = R_f \left(\frac{V_1}{2R} + \frac{V_2}{4R} + \frac{V_3}{8R} + \frac{V_4}{16R} \right)$$

(b) If $R_f = 12 \text{ k}\Omega$ and $R = 10 \text{ k}\Omega$, find $|V_o|$ for $[V_1 V_2 V_3 V_4] = [1011]$ and $[V_1 V_2 V_3 V_4] = [0101]$.

Figure. 5.103:



Step-by-step solution

Step 1 of 4

(a)

Consider a six-bit digital convertor.

The expression of the output voltage of the six-bit digital convertor is,

$$V_o = - \left[\left(\frac{R_f}{R_1} \right) V_1 + \left(\frac{R_f}{R_2} \right) V_2 + \left(\frac{R_f}{R_3} \right) V_3 + \left(\frac{R_f}{R_4} \right) V_4 + \left(\frac{R_f}{R_5} \right) V_5 + \left(\frac{R_f}{R_6} \right) V_6 \right]$$

Make the bits are weights according to the magnitude of their place value, by descending value

of $\frac{R_f}{R_n}$ so that each lesser bit has half the weight of the next higher.

Choose $R_1 = 2^0 R$, $R_2 = 2^1 R$, $R_3 = 2^2 R$, $R_4 = 2^3 R$, $R_5 = 2^4 R$ and $R_6 = 2^5 R$.

Substitute corresponding assumptions in the equation.

$$V_o = - \left[\left(\frac{R_f}{2^0 R} \right) V_1 + \left(\frac{R_f}{2^1 R} \right) V_2 + \left(\frac{R_f}{2^2 R} \right) V_3 + \left(\frac{R_f}{2^3 R} \right) V_4 + \left(\frac{R_f}{2^4 R} \right) V_5 + \left(\frac{R_f}{2^5 R} \right) V_6 \right]$$
$$V_o = - \left[\left(\frac{R_f}{R} \right) V_1 + \left(\frac{R_f}{2R} \right) V_2 + \left(\frac{R_f}{4R} \right) V_3 + \left(\frac{R_f}{8R} \right) V_4 + \left(\frac{R_f}{16R} \right) V_5 + \left(\frac{R_f}{32R} \right) V_6 \right]$$

Assume $R_f = 10 \text{ k}\Omega$ and $R = 10 \text{ k}\Omega$.

$$V_o = - \left[\left(\frac{10}{10} \right) V_1 + \left(\frac{10}{2(10)} \right) V_2 + \left(\frac{10}{4(10)} \right) V_3 + \left(\frac{10}{8(10)} \right) V_4 + \left(\frac{10}{16(10)} \right) V_5 + \left(\frac{10}{32(10)} \right) V_6 \right]$$
$$= - \left(V_1 + \frac{V_2}{2} + \frac{V_3}{4} + \frac{V_4}{8} + \frac{V_5}{16} + \frac{V_6}{32} \right)$$

Step 2 of 4

The magnitude of the output voltage is,

$$|V_o| = V_1 + \frac{V_2}{2} + \frac{V_3}{4} + \frac{V_4}{8} + \frac{V_5}{16} + \frac{V_6}{32}$$
$$= V_1 + 0.5V_2 + 0.25V_3 + 0.125V_4 + 0.0625V_5 + 0.03125V_6$$

Choose $V_1 = 1$, $V_2 = 0$, $V_3 = 0$, $V_4 = 1$, $V_5 = 1$ and $V_6 = 0$.

$$|V_o| = 1 + 0 + 0 + 0.125 + 0.0625 + 0$$
$$= 1.1875 \text{ V}$$

Thus, the value of $[V_1 V_2 V_3 V_4 V_5 V_6]$ is 100110.

Step 3 of 4

(b)

Write the expression of the magnitude of the output voltage.

$$\begin{aligned} |V_o| &= V_1 + \frac{V_2}{2} + \frac{V_3}{4} + \frac{V_4}{8} + \frac{V_5}{16} + \frac{V_6}{32} \\ &= V_1 + 0.5V_2 + 0.25V_3 + 0.125V_4 + 0.0625V_5 + 0.03125V_6 \end{aligned}$$

Substitute $[011011]$ for $[V_1V_2V_3V_4V_5V_6]$.

$$\begin{aligned} |V_o| &= 0 + 0.5(1) + 0.25(1) + 0.125(0) + 0.0625(1) + 0.03125(1) \\ &= 0.5 + 0.25 + 0.0625 + 0.03125 \\ &= 0.84375 \end{aligned}$$

Thus, the output voltage V_o value for $[011011]$ is $\boxed{0.84375 \text{ V}}$.

Step 4 of 4

(c)

Write the expression of the magnitude of the output voltage.

$$\begin{aligned} |V_o| &= V_1 + \frac{V_2}{2} + \frac{V_3}{4} + \frac{V_4}{8} + \frac{V_5}{16} + \frac{V_6}{32} \\ &= V_1 + 0.5V_2 + 0.25V_3 + 0.125V_4 + 0.0625V_5 + 0.03125V_6 \end{aligned}$$

Substitute $[111111]$ for $[V_1V_2V_3V_4V_5V_6]$ to obtain the maximum value for V_o .

$$\begin{aligned} |V_o| &= 1 + 0.5(1) + 0.25(1) + 0.125(1) + 0.0625(1) + 0.03125(1) \\ &= 1 + 0.5 + 0.25 + 0.125 + 0.0625 + 0.03125 \\ &= 1.96875 \text{ V} \end{aligned}$$

Thus, the maximum output voltage V_o value for $[011011]$ is $\boxed{1.96875 \text{ V}}$.